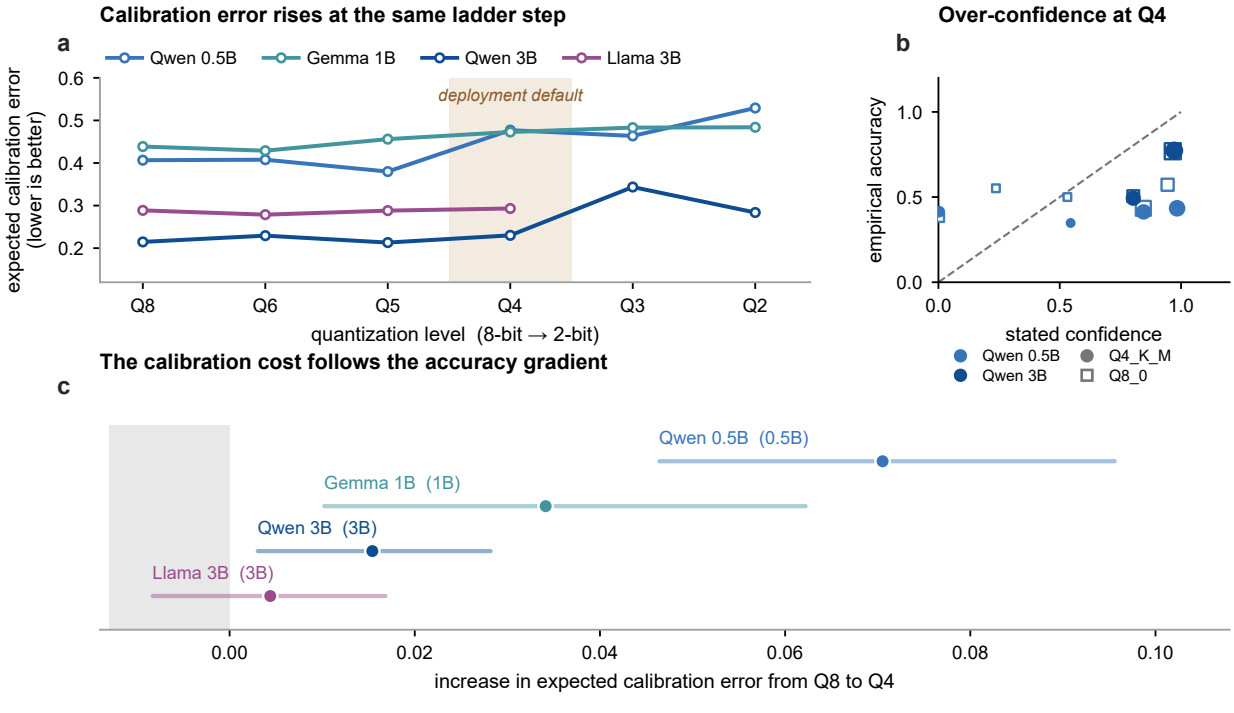

Calibration error rises at the same ladder step
a
Qwen 0.5B
Gemma 1B
Qwen 3B
Llama 3B
deployment default
expected calibration error
(lower is better)
0.6
0.5
0.4
0.3
0.2
Q8
Q6
Q5
Q4
Q3
Q2
quantization level (8-bit → 2-bit)
Over-confidence at Q4
b
empirical accuracy
1.0
0.5
0.0
0.0
0.5
1.0
stated confidence
Qwen 0.5B
Qwen 3B
Q4_K_M
Q8_0
The calibration cost follows the accuracy gradient
c
Qwen 0.5B (0.5B)
Gemma 1B (1B)
Qwen 3B (3B)
Llama 3B (3B)
0.00
0.02
0.04
0.06
0.08
0.10
increase in expected calibration error from Q8 to Q4